

Protein Requirements

in Inherited Metabolic Diseases

Abbreviations · Full Disease Names · Age-Based Protein Tables

Condensed reference — Chapter 6, Yannicelli (2022)

Nutrition Management of Inherited Metabolic Diseases, 2nd Edition

Bernstein LE · Rohr F · van Calcar S (Eds.) — Springer 2022

Disease Abbreviations & Full Names

Complete list of abbreviations used in this reference:

Abbreviation	Full Disease Name	Deficient Enzyme / Gene
PKU	Phenylketonuria	Phenylalanine hydroxylase (PAH)
HPA	Hyperphenylalaninemia (mild PKU)	PAH — partial deficiency
MSUD	Maple Syrup Urine Disease	Branched-chain keto-acid dehydrogenase (BCKD)
HCU	Homocystinuria (CBS type)	Cystathionine beta-synthase (CBS)
HT-1	Hereditary Tyrosinemia Type 1	Fumarylacetoacetate hydrolase (FAH)
HT-2	Hereditary Tyrosinemia Type 2	Tyrosine aminotransferase (TAT)
HT-3	Hereditary Tyrosinemia Type 3	4-hydroxyphenylpyruvate dioxygenase (HPD)
OTC	Ornithine Transcarbamylase Deficiency	OTC enzyme (X-linked)
CPS1	Carbamoyl Phosphate Synthetase 1 Deficiency	CPS1
ASS / CTLN1	Citrullinemia Type 1	Argininosuccinate synthetase (ASS1)
ASL	Argininosuccinic Aciduria	Argininosuccinate lyase (ASL)
ARG1	Arginase 1 Deficiency / Hyperargininemia	Arginase 1 (ARG1)
NAGS	N-Acetylglutamate Synthase Deficiency	NAGS
UCD	Urea Cycle Disorders (collective)	Any UCD enzyme above
PA	Propionic Acidemia	Propionyl-CoA carboxylase (PCCA/PCCB)
MMA	Methylmalonic Acidemia	Methylmalonyl-CoA mutase (MUT) or Cbl defect
GA-1	Glutaric Acidemia Type 1	Glutaryl-CoA dehydrogenase (GCDH)
IVA	Isovaleric Acidemia	Isovaleryl-CoA dehydrogenase (IVD)
VLCADD	Very-Long-Chain Acyl-CoA Dehydrogenase Deficiency	ACADVL
LCHAD	Long-Chain 3-Hydroxyacyl-CoA Dehydrogenase Def.	HADHA
TFP	Trifunctional Protein Deficiency	HADHA / HADHB
MCADD	Medium-Chain Acyl-CoA Dehydrogenase Deficiency	ACADM
SCADD	Short-Chain Acyl-CoA Dehydrogenase Deficiency	ACADS
FAOD	Fatty Acid Oxidation Disorders (collective)	Any FA oxidation enzyme above
LC-FAOD	Long-Chain Fatty Acid Oxidation Disorders	VLCADD / LCHAD / TFP

Abbreviation	Full Disease Name	Deficient Enzyme / Gene
GALT	Classical Galactosemia	Gal-1-P uridyltransferase (GALT)
GSD-I	Glycogen Storage Disease Type I (von Gierke)	Glucose-6-phosphatase (G6PC)
GSD-Ib	Glycogen Storage Disease Type Ib	G6P translocase (SLC37A4)
GSD-IIIa	Glycogen Storage Disease Type IIIa	Glycogen debranching enzyme (AGL)

Why IMD Patients Require More Protein Than the RDA

Key finding from Chapter 6 (Yannicelli 2022):

Free amino acids (in Medical Food) are absorbed faster, oxidised at higher rates, and retained less efficiently than intact protein. A 20-25% correction above the RDA is required to achieve equivalent nitrogen balance.

Factor	Intact Protein (normal diet)	Free AA Medical Food (IMD)
Absorption speed	Slow — sustained amino acid peak	Fast — rapid peak & higher oxidation
Nitrogen retention	High	Lower — ~20-25% more protein needed
RDA adequacy	Meets 97-98% of healthy individuals	Underestimates need — use 120-140% RDA
IAAO finding	0.95 g/kg/day (adult RDA)	Children up to 1.55 g/kg/day (95th CI)
Distribution	Flexible timing	3+ equal doses/day to maximise synthesis

CORE RULE — Chapter 6

Total protein = 120-140% of DRI/RDA when the majority comes from amino-acid-based Medical Food. Divide Medical Food across ≥ 3 meals/day — a single bolus dose leads to higher oxidation and lower retention.

Protein:Energy Ratio — Key thresholds:

Context	Protein Target
Standard infant formula minimum (Infant Formula Act 1980)	1.8 g protein / 100 kcal (7.2% of energy)
AA-based / hydrolysate infant formulas	> 1.8 g / 100 kcal (adjusted for digestibility)
PKU children 6-18 yr (IAAO method)	1.85 g total protein / kg / day (~19% of energy)
PKU growth target P:E ratio	3.0-4.5 g protein / 100 kcal
PKU total protein — infants/toddlers	1.5-2.6 g/kg/day (intact protein >0.5 g/kg/day)
Catch-up growth (any IMD with growth failure)	9-12% of total energy as protein (higher than DRI)
Organic acidemias — Medical Food proportion	21% (mild PA) to 80% (severe PKU) of total protein

Protein Requirements by Disease — Stable State

PKU | Phenylketonuria

Age	PHE (mg/day)	TYR (mg/day)	Total Protein (g/kg)
0-3 mo	130-430	1100-1300	3.0-3.5
3-6 mo	135-400	1400-2100	3.0-3.5
6-9 mo	145-370	2500-3000	2.5-3.0
9-12 mo	135-330	2500-3000	2.5-3.0
1-4 yr	200-320	2800-3500	>= 30 g/day total
> 4 yr-adult	200-1100	4000-6000	120-140% RDA for age

Source: GMDI/SERN Table 6.2. Protein from PHE-free AA-based Medical Food.

MSUD | Maple Syrup Urine Disease

Age	LEU (mg/kg)	ILE (mg/kg)	VAL (mg/kg)	Protein (g/kg)	Energy (kcal/kg)
0-6 mo	40-100	30-90	40-95	2.5-3.5	95-145
7-12 mo	40-75	30-70	30-80	2.5-3.0	80-135
1-3 yr	40-70	20-70	30-70	1.5-2.5	80-130
4-8 yr	35-65	20-30	30-50	1.3-2.0	50-120
9-13 yr	30-60	20-30	25-40	1.2-1.8	40-90
14-18 yr	15-50	10-30	15-30	1.2-1.8	35-70
>= 19 yr	15-50	10-30	15-30	1.1-1.7	35-45

UCD | Urea Cycle Disorders (OTC - CPS1 - ASS - ASL - ARG1 - NAGS)

Age (yr)	Total Protein (g/kg/day)	From EAA Medical Food (g/kg/day)	Intact Protein (g/kg/day)
1-7	1.0-1.2	0.3-0.7	0.7-0.8
7-19	0.8-1.4	0.4-0.7	0.3-1.0
Adults	0.8-1.0+	20-30% of total	Remainder

EAA supplement = 20-30% of total protein. Never restrict protein > 24 h.

Protein Requirements by Disease — Stable State (cont.)

HCU | Homocystinuria (CBS deficiency)

Age	Methionine (mg/kg/day)	Cystine (mg/kg/day)	Total Protein
Infant 0-12 mo	25-50	50-100	100-140% DRI
4-7 yr	7-22	50-80	100-140% DRI
7-11 yr	7-22	30-50	100-140% DRI
Adults	Individualise	Individualise	100-140% DRI

Intact protein = 60-100% of DRI. Total 120-140% DRI if mostly Medical Food.

HT-1 / HT-2 / HT-3 | Hereditary Tyrosinemia (all types)

Parameter	Value
PHE at diet start	185-550 mg/day
TYR at diet start	95-275 mg/day
Total protein	Normal DRI — via PHE/TYR-free Medical Food + limited intact protein
Plasma TYR target	< 600 umol/L (on NTBC)
NTBC dose (HT-1)	1 mg/kg/day in 2 divided doses

GA-1 | Glutaric Acidemia Type 1

Age	Total Protein (g/kg)	Lysine (mg/kg)	L-Carnitine (mg/kg)
0-180 days	2.5-3.5	70-100	50-100
6-24 mo	1.5-3.0	60-80	50-100
2-6 yr	1.3-2.5	40-60	50-100
> 6 yr	DRI	Protein-controlled	50

PA / MMA | Propionic Acidemia & Methylmalonic Acidemia

Parameter	Recommendation
Total protein	100-120% of DRI — from propiogenic-AA-free Medical Food + intact protein
Intact protein	Titrate to 100% DRI for age — do NOT over-restrict
Medical Food share	21-50% of total protein (avoid overprescription)
Restricted AAs	Val, Met, Ile, Thr (propiogenic) — from Medical Food devoid of these
L-Carnitine	100-300 mg/kg/day in 2-4 divided doses

Protein Requirements by Disease — Stable State (cont.)

FAOD | Fatty Acid Oxidation Disorders

Parameter	Value
Protein target	25% of total energy — preserves lean mass, improves lipid profile
Protein source	Normal intact protein — no specific amino acid restriction
Medical Food	Not required (no AA restriction; focus is on fat modification)
Clinical note	25% protein energy vs 12% is superior for body composition (Chapter 23)

GALT | Classical Galactosemia

Parameter	Value
Total protein	100% DRI — normal requirements, no protein restriction
Protein source	Galactose/lactose-free sources: soy, meat, fish, eggs, legumes
Infant formula	Soy protein isolate formula or elemental amino acid formula
Restriction	Galactose and lactose only — protein intake unrestricted

GSD-I / GSD-IIIa | Glycogen Storage Disease

Type	Protein Target	Notes
GSD-I	10-15% of total energy	Carbohydrate is the primary fuel. Excess protein not beneficial.
GSD-IIIa	20-30% of energy / 3-4 g/kg/day	High protein supports gluconeogenesis and muscle function. Titrate upward.

Protein During Acute Metabolic Crisis — All IMD

Phase	Action	Max Duration	Rationale
Initial crisis	Remove or severely restrict all protein	Max 24 h (48-72 h absolute limit)	Reduce toxic metabolite production
Transition	Restart at 0.5 g/kg/day; advance by 0.25 g/kg/day every 24 h	As soon as stable	Catabolism worsens toxicity more than dietary protein
Energy during restriction	IV dextrose 8-12 mg/kg/min + Intralipid 2-3 g/kg/day	Throughout protein-free phase	Prevent catabolism — energy deficit is primary driver
AA Medical Food	Continue (without offending AAs)	Throughout crisis	Maintains nitrogen balance; reduces endogenous breakdown

KEY PRINCIPLE — Chapter 6 Summary

Total protein requirements in IMD are HIGHER than DRI when the majority comes from amino-acid-based Medical Food. Free AAs are oxidised faster than intact protein, requiring a 20-25% correction. Distribute Medical Food in 3+ equal doses throughout the day to maximise nitrogen retention and protein synthesis.

Quick Reference Summary — All Diseases

Disease	Total Protein Target	Intact Protein	Medical Food %	Restricted AAs
PKU	120-140% DRI (infant: 3.0-3.5 g/kg)	Titrate to blood PHE	~80%	PHE
MSUD	2.5-3.5 g/kg (infant) to DRI (adult)	Titrate to blood LEU	80-90% classical	BCAA (LEU·ILE·VAL)
OTC / CPS1	1.0-1.2 g/kg (1-7 yr)	0.7-0.8 g/kg	20-30%	Total N limited
ASS / ASL	1.0-1.4 g/kg (7-19 yr)	0.3-1.0 g/kg	20-30%	Total N limited
HCU	100-140% DRI	60-100% of DRI	20-40%	Met
HT-1/2/3	Normal DRI	Limited by PHE+TYR tolerance	Variable	PHE + TYR
GA-1	2.5-3.5 g/kg (infant)	Titrate to blood Lys	Majority	LYS (+TRP)
PA / MMA	100-120% DRI	50-80% of DRI	20-50%	Val·Met·Ile·Thr
FAOD	25% of energy	Normal — no restriction	None needed	None (fat only)
GALT	100% DRI	Normal lactose-free sources	Soy (infants)	None
GSD-I	10-15% of energy	Lean sources — normal quality	None needed	None
GSD-IIIa	3-4 g/kg/day	High — gluconeogenesis support	Optional supp.	None

For educational purposes only. Individualise based on plasma amino acids, growth, and clinical status.
Source: Nutrition Management of Inherited Metabolic Diseases, 2nd Ed. — Bernstein, Rohr, van Calcar — Springer 2022